



RICH PLUGIN

Project By Alexandre Younes, Georgios Abboud

Comity Dr. Marcel Karam (**Advisor**)
Dr. Nabil Fares, Dr. Carine Kassis, Dr. Ahmad Faour

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Many sessions for
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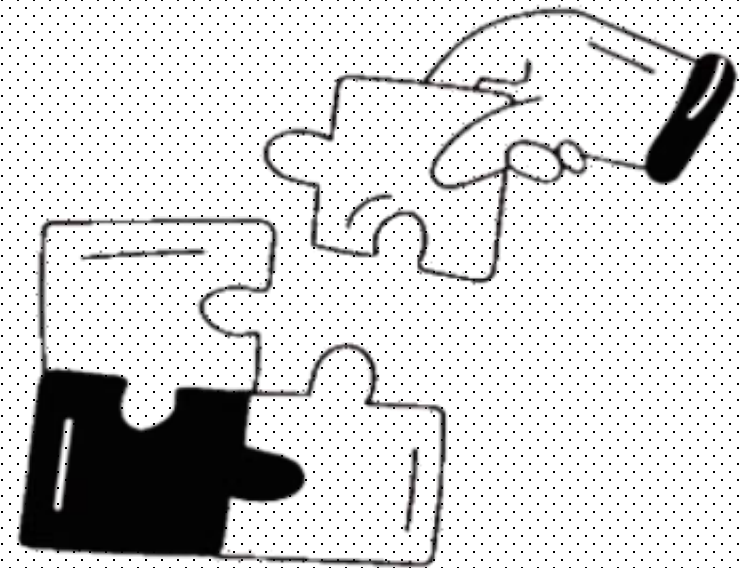
Related Work:

#16

THE PROBLEM

Traditional Systems = One-Fit-All

- Fixed problems → Fixed solutions
(e.g., *Login issue?* → Show generic tooltip)
- Every user treated the same, regardless of experience, intent, or struggle.
- No adaptation to behavior, context, or past failures.



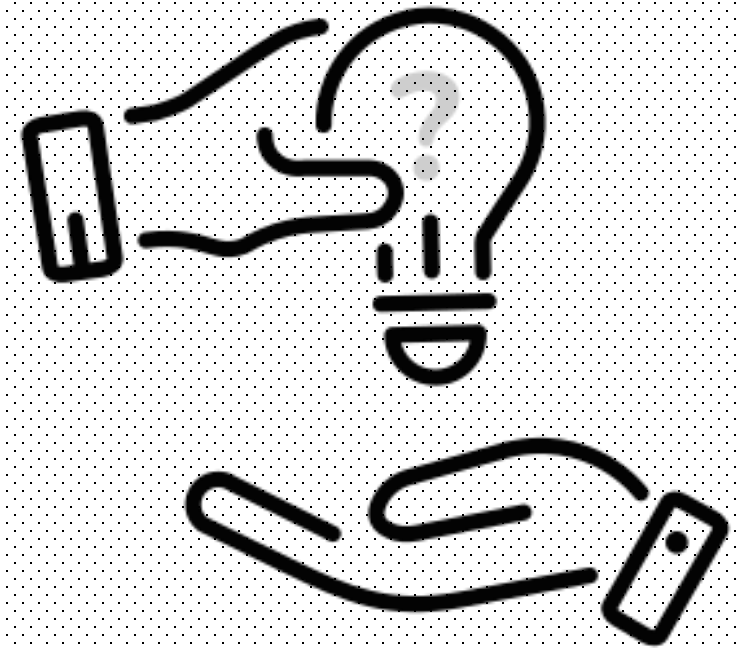
THE SOLUTION

Our Approach = Adaptive & Personalized

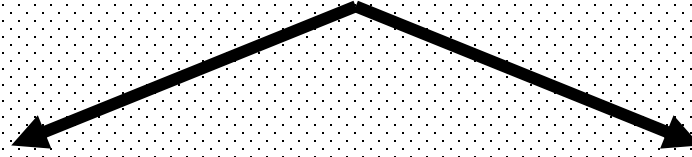
Analyze **individual behavior** in real time.
Understand **why** a user is struggling, not just **what** they're doing.

Dynamically generate support:

- Autofocus inputs
- Suggest intelligent completions
- Modify DOM to guide the user
- Trigger contextual help only when needed

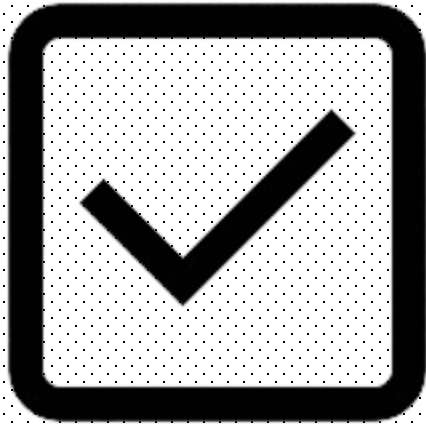


SYSTEM / USER EVENTS (web app)



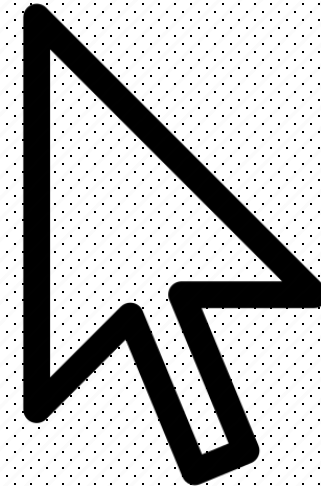
Transitional event
(alters dom)

Ex: checkbox, submit



Non-Transitional event
(does not alters dom)

Ex: dead clicks, hover , Scroll,



Example of user behavior

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Category	Description	Core Events	Insights	Actions
Decision Uncertainty & Hesitation [2], [3]	Signals of user indecision, confusion, or second-guessing	• Pointer idle • Pause before submit • Form dwell time • Oscillating hovers • Form filled then cleared • Modal opened then canceled • Keydown without submit • Frequent Escape/Backspace use	• Cognitive load peaks when choices are unclear [3] • Privacy/trust concerns at commit points [2]	• Simplify form fields and reduce steps [3] • Add contextual tooltips and inline help [2] • Prompt reassurance (“Everything look okay?”)
Frustration & UX Breakdown Indicators [5], [6]	High-urgency signals of breakdowns or abandonment risk	• Repeated clicks (“rage clicks”) on same element • Failed swipe/gesture attempts • Interaction with hidden/disabled elements	• Rage clicks correlate strongly with user frustration [5] • Hidden/disabled controls spike abandonment [6]	• Trigger live chat or contextual help on frustration signals [5] • Prioritize fixing unresponsive UI components [6]

Capturing user behavior

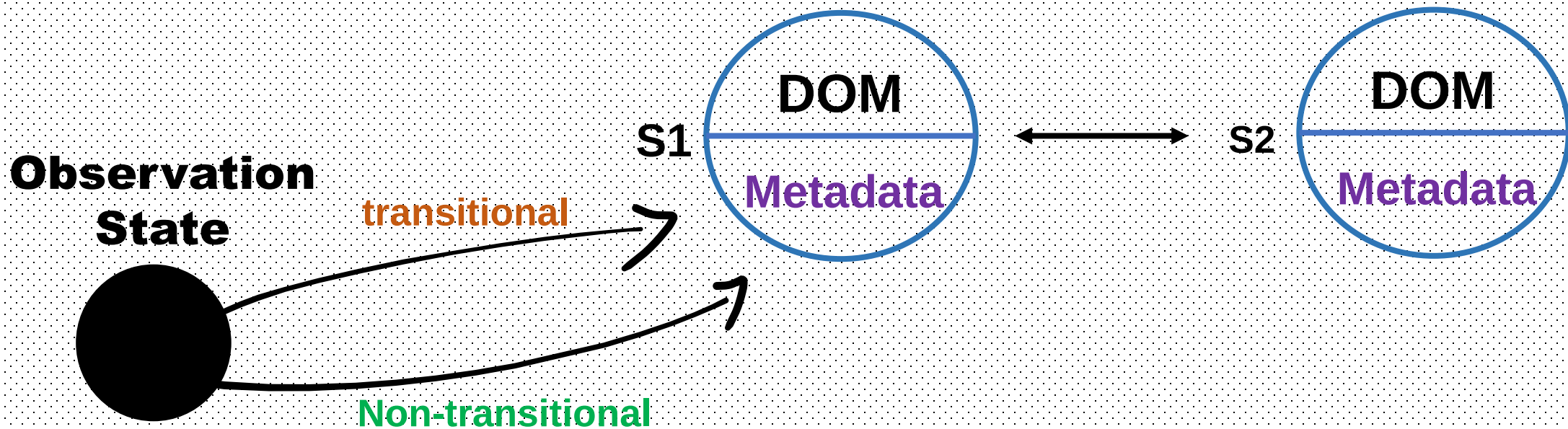
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Using **FSM** (where each state capture dom with some meta data)

Our FSM:

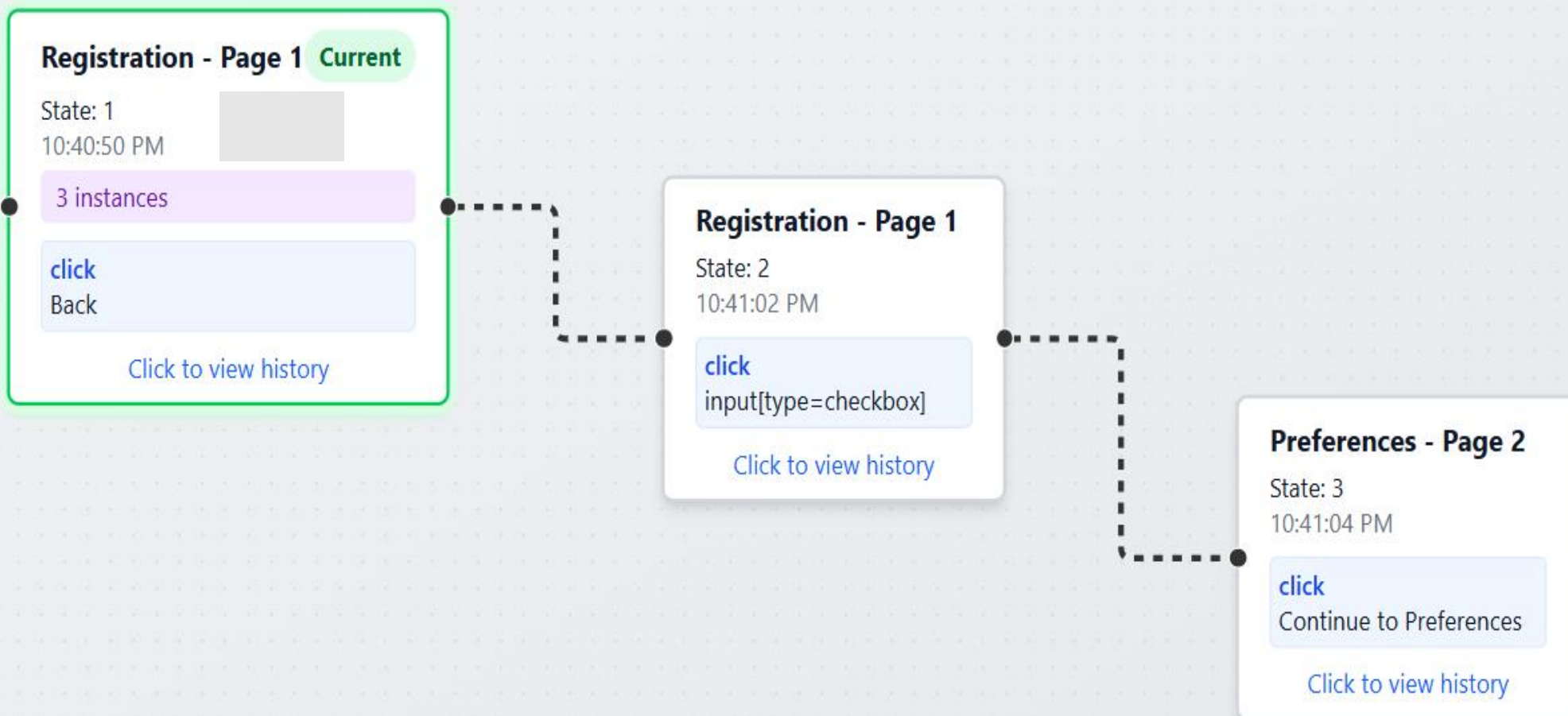
- no initial state (different loading time for different user)
- Observational state (capture non transnational)

How FSM is constructed



Session example

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Non-Transitional Metrics

Summary of quantitative measurements of user behavior during this state

Time Metrics		Mouse Metrics		Keyboard Metrics	
Dwell Time Before Action	22.84s	Total Clicks	25	Total Keystrokes	31
Longest Idle Period	92.50s	Total Hover Time	92.48s		
Total Hover Time	92.48s	Total Mouse Distance	12275px		
Total Idle Time	460.30s				

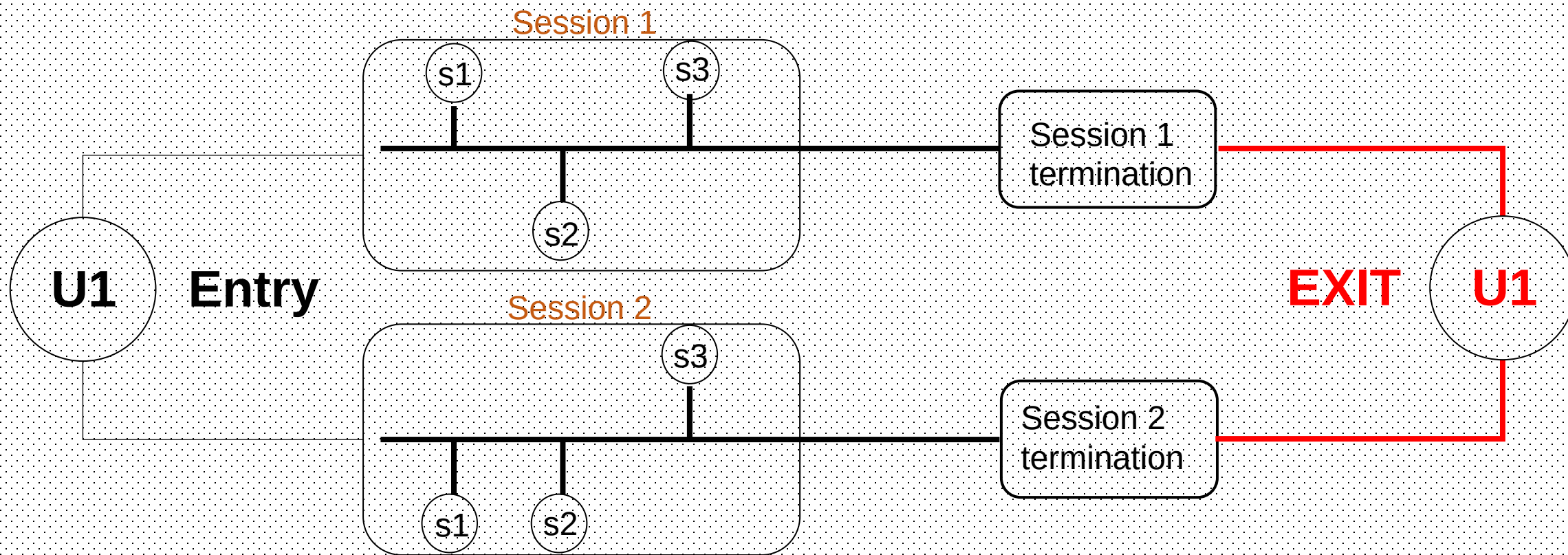
Non-Transitional Events

Detailed record of specific user interactions that don't cause page navigation

104 Hover Events100.02s total hover time

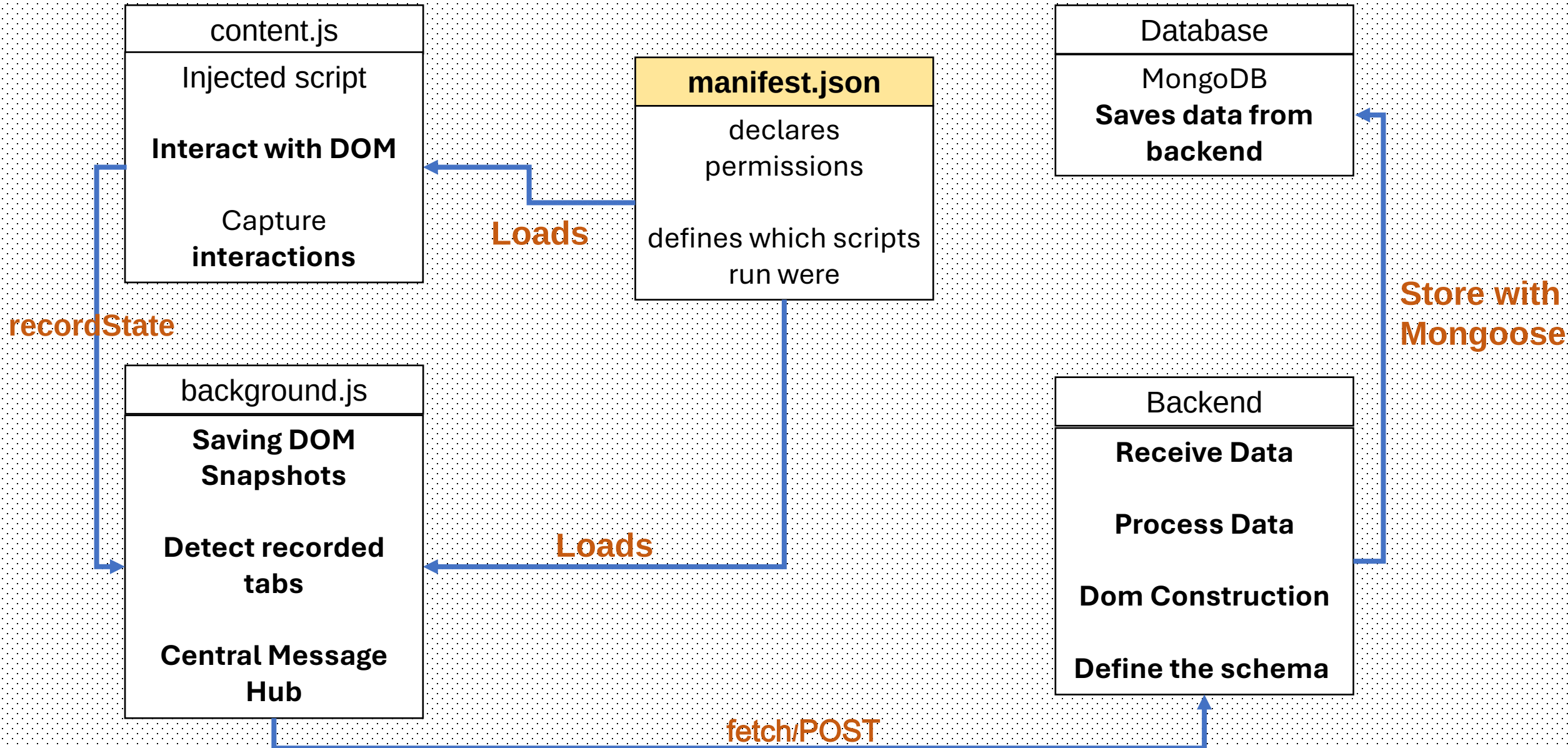
Element	Selector	Duration	Timestamp
label	#registrationForm > div.form-group > label:nth-child...	0.14s	10:40:50 PM
input	#fullName	0.58s	10:40:50 PM
div	html > body > div.container:nth-child(1)	0.06s	10:40:51 PM
body	html > body:nth-child(2)	0.22s	10:40:51 PM
div	html > body > div.container:nth-child(1)	0.07s	10:40:51 PM
h1	html > body > div.container:nth-child(1) > h1:nth-child(1)	5.53s	10:40:51 PM

Many sessions for one user



Architecture

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Transitional model

```
_id: ObjectId('682c09b0a72e4cdc13035d60')
stateId: "state_1_1747716528177"
sessionId: "session_1747716528058"
userId: "user_148075a0-61bb-49d8-be28-39bba914fed9"
url: "http://127.0.0.1:5500/index.html"
pathname: "/index.html"
timestamp: 2025-05-20T04:48:48.174+00:00
isNewState: true
stateNumber: 1
hash: "z368zn_indexhtml"
previousStateId: null
previousHash: null
timeSincePreviousState: 0
dom: "<html lang='en'><head><style>body {transition
▸ metrics: Object
  title: "Registration - Page 1"
  interactionInfo: null
▸ loadingInfo: Object
▸ mutationInfo: Object
  createdAt: 2025-05-20T04:48:48.190+00:00
  updatedAt: 2025-05-20T04:48:48.190+00:00
  __v: 0
```

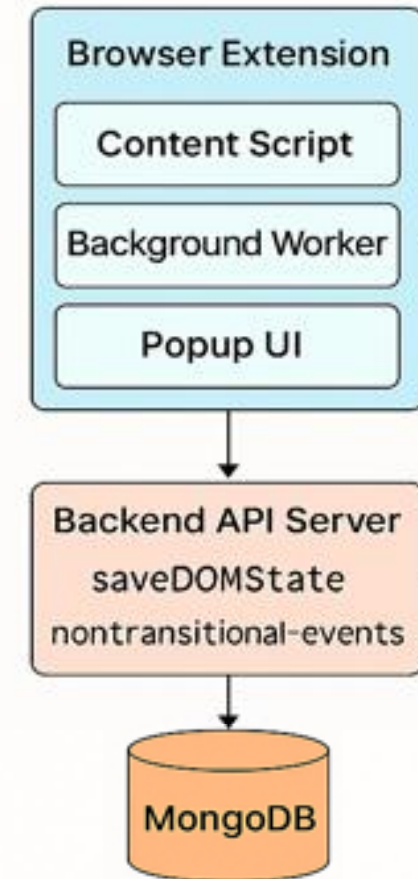
Non - Transitional model

```
_id: ObjectId('682c09b468a0ceb600a35ecc')
sessionId: "session_1747716528058"
stateId: "state_1_1747716528177"
userId: "user_148075a0-61bb-49d8-be28-39bba914fed9"
__v: 0
createdAt: 2025-05-20T04:48:52.062+00:00
▾ events: Object
  ▸ allKeyPresses: Array (33)
  ▸ copyText: Array (empty)
  ▸ deadClicks: Array (1)
  ▸ dropdownToggle: Array (empty)
  ▸ escapeBackspace: Array (19)
  ▸ hover: Array (77)
  ▸ inactivity: Array (2)
  ▸ inputContent: Array (5)
  ▸ inputFieldIdle: Array (5)
  ▸ keyTypingCadence: Array (28)
  ▸ keydownWithoutSubmit: Array (empty)
  ▸ mousemove: Object
  ▸ oscillatingHovers: Array (12)
  ▸ pasteWithoutTyping: Array (empty)
  ▸ repeatedClicks: Array (1)
  ▸ repeatedInputs: Array (1)
  ▸ scrollEvents: Array (empty)
  ▸ tabNavigation: Array (empty)
  lastUpdated: 2025-05-20T04:50:40.828+00:00
  ▸ metrics: Object
```

The Flow

Complete End-to-End Flow

- Browser Extension captures user behavior in real time.
 - Content Script: watches DOM mutations, clicks, and hovers.
 - Background Worker: compresses, deduplicates, and batches data
 - Popup UI: lets user start/stop session
- Backend API Server ingests and validates:
 - saveDOMState → saves snapshots to MongoDB
 - Powered by Express and the ServiceManager



AI analysis

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AI Insight Pipeline (LLM-Powered)

Session Data Ingested

Aggregates recorded user sessions (DOM + behavior) from MongoDB.

Stage 1: Transitional Analysis

- Deep analysis of user transitional event

Stage 2: Non - Transitional Analysis

- Deep analysis of user non- transitional event

Stage 3: Pain Point Q&A

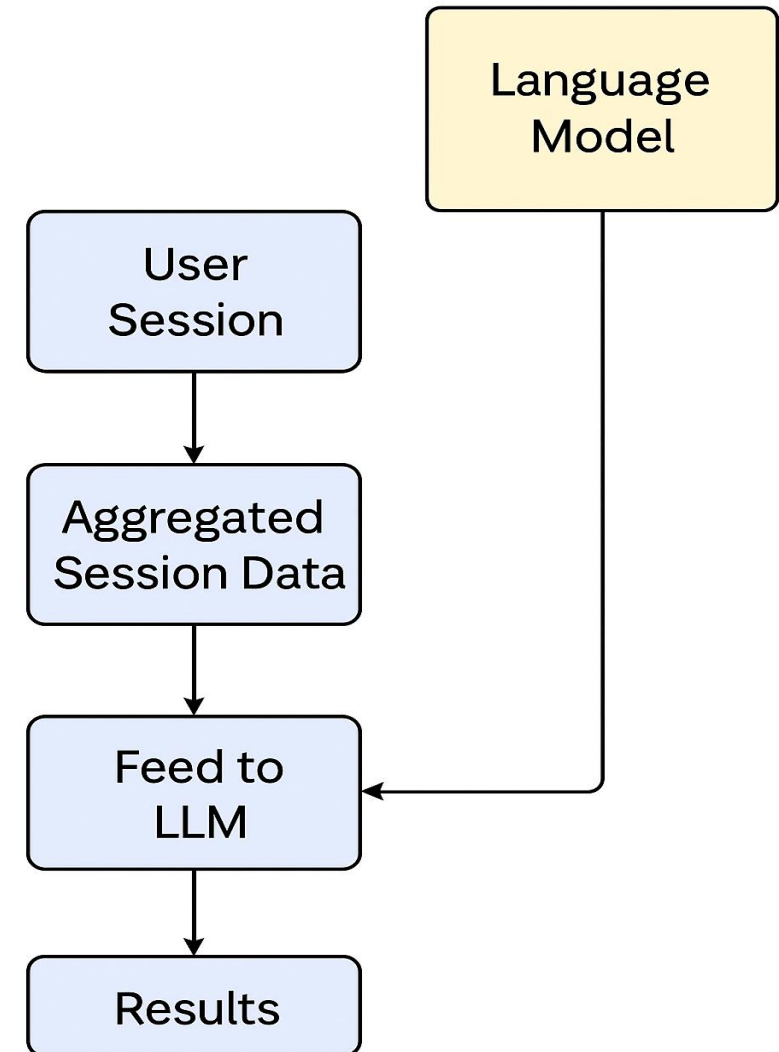
- LLM pinpoints user struggle moments with reasoning and evidence.

Stage 4: UX Help Generation

- LLM proposes precise interventions (e.g., auto-fill, tooltip, skip step).

Insights Saved + Pushed

- Results are stored and pushed to dashboards via Socket.IO.



Future Goal

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Down the line, we plan to **dynamically generate real-time support**

meaning, as soon as our extension detects user frustration (like erratic hovers or dead clicks), we'll trigger live guidance: maybe a tooltip or a UI change suggestion powered by AI.”

● Related Work:

1. **Traditional DOM-Based** Analytics Frameworks
2. **Machine Learning** for DOM Structure Analysis
3. **Visual-DOM Correlation** Techniques
4. **Deep Learning Approaches** for DOM Analytics
5. **Performance-Oriented** DOM Analytics
6. **Behavioral Pattern Extraction**
7. **Cross-Device** DOM Analytics
8. **Accessibility-Oriented** DOM Analytics
9. **Security and Privacy** Considerations
10. **Real-time** DOM Analytics

